

Prem Kumar Mahanty

GEN AI Developer

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Education

- **VIGNAN'S INSTITUTE OF INFORMATION TECHNOLOGY (B.Tech. Computer Science)** 2024–2028
CGPA: 8.6/10
- **INDIAN INSTITUTE OF TECHNOLOGY, Madras (B.Sc. Data Science and AI)** 2024–2028
CGPA: 7.79/10

Experience

LawVriksh — GenAI Intern

Aug 2025 – Present

- Built a full **compliance-aware LLM pipeline** with deterministic prompting, source-grounded generation, and multi-layer verification, reducing hallucination rates by 72% across legal drafting workflows.
- Engineered an **automated fact-checking engine** using retrieval-augmented verification, cosine-similarity scoring, and cross-document conflict detection, improving factual accuracy by 68%.
- Developed a **reference-manager-driven citation architecture** with summary embeddings, weighted keyword ranking, and auto-generated citation blocks, lowering retrieval latency by 55%.
- Implemented a **multi-agent system** (ingestion, retrieval, analysis, generation, compliance agents) using MCP servers, enabling controllable tool use, traceable outputs, and safe agentic reasoning.
- Designed a **real-time gap analysis engine** that identifies missing arguments, supporting principles, and counter-reasoning using hybrid RAG + rule-based logic, boosting document completeness by 40%.
- Integrated **LLM observability modules** to log prompt-output lineage, ensure auditability, and support compliance teams in reviewing AI-generated drafts.
- Enhanced legal drafting UX with **context-aware suggestion generation** (research-grounded vs. context-only modes), achieving 2.3× faster drafting throughput for end-users.

Somtax Solutions — AI Intern

Aug 2025 – December 2025

- Designed and deployed multimodal AI agents and a citation engine using custom MCP servers & APIs, achieving seamless data integration and automation while cutting operational costs by 40%.
- Orchestrated a scalable microservices architecture with Django & Docker, implementing CI/CD pipelines that reduced deployment time by 40% and improved system reliability by 25% while leading a team of 5 engineers.
- Pioneered a next-generation recommendation engine with Graph Neural Networks (GNNs) + Faiss, replacing legacy systems to boost processing speed by 60% and optimize cost efficiency.

Projects

- **Compliance-Aware LLM Writing Verification Engine** *Python, FastAPI, LangChain, FAISS, Gemini/GPT, Redis, Async IO*
 - Designed and deployed a multi-stage retrieval and compliance pipeline that reduced hallucinated statements by **72%** through deterministic prompting, reference validation, and hybrid RAG scoring.
 - Built an optimized embedding + vector search layer using FAISS and Redis caching, decreasing retrieval latency by **40–55%** across high-volume requests.
 - Implemented asynchronous FastAPI endpoints and batched inference flows, improving end-to-end response time from **1.9s → 0.8s**.
 - Developed an automated citation-insertion and fact-violation detection module that increased correct citation placement accuracy by **65%**.
 - Created a context-window reduction algorithm that shrank token consumption by **38%** while preserving relevance via ranked chunking and adaptive windowing.
 - Oversaw system profiling, regression tests, and latency audits, enabling stable performance at **10k+** monthly compliance-check executions.
- **E-Commerce Website** *React, Django, MySQL,*
 - Designed and developed a fully functional e-commerce website with responsive UI.
 - Integrated the e-commerce site with third-party APIs for shipping, payment, and tax calculation, automating order fulfillment processes; reduced manual order processing time by 4 hours weekly.
 - Utilized Django for backend logic and MySQL for data storage.
- **Swaply** *Tailwind CSS, Django, PostgreSQL, JS, MCP Servers, GNN's, Agentic Architect* [Live](#)
 - Developed a coupon trading platform with Django and PostgreSQL, supporting secure transactions, chat, and wallet systems.
 - Implemented real-time screen sharing and live coupon verification using WebRTC and Django Channels with WebSockets.
 - Designed a mobile-first responsive interface using Tailwind CSS, enabling seamless user experience and profile management.

Skills

- **Programming Languages and Databases:** SQL, Java, MongoDB, Python, PHP, JavaScript, MySQL
- **Web tools and Frameworks:** HTML, CSS & ReduxToolkit, Django, Bootstrap, React
- "Led a team of four to develop a YouTube clone, demonstrating excellent teamwork."
- **AI/ML & Data Science:** PyTorch, TensorFlow, Scikit-learn
- **Recommendation Systems:** Graph Neural Networks (GNNs), Faiss (vector similarity search)
- **DevOps & Infrastructure:** Docker, CI/CD Pipelines, Microservices Architecture, Git/GitHub
- **APIs & Backend Development:** Django REST Framework, Custom MCP Servers, REST API design

Certifications and Achievements

- **Hacakthon winner of WOW GOOGLE DEVELOPERS GROUP 2025 and finalist in the Gen E-Summit AI Ideathon**